Logic, Language and Philosophy
Course Schedule

version 2

2nd August 2025

1 Preliminary

Basic parameters:

- The course runs for 15 weeks
- There are 2 exams, one intermediate, one final
- Each section has two meetings, of 95 minutes each
- The course will be taught in English

Course material:

- Liu Fenrong and Martin Stokhof, *Logic, Language and Philosophy. An Introduction to Standard Logic*, CSLI Press, Stanford, 2024 (Liu and Stokhof, 2024)
- Papers and book chapters from Pascal Engel (Engel, 2006), Gottlob Frege (Frege, 1879, 1948), Gilbert Harman (Harman, 1986), Saul Kripke (Kripke, 1971), Arthur Prior (Prior, 1960), W.V.O. Quine (Quine, 1948), Ruy de Quieroz (de Queiroz, 2008), Bertrand Russell (Russell, 1914, 1905), David Sanford (Sanford, 1989), Peter Strawson (Strawson, 1950), Peter Suber (Suber, 1997), Alfred Tarski (Tarski, 1944).

The textbook present the logical material along with introductions to the papers that are used to illustrate the philosophical relevance of various logical concepts and techniques. There is a Chinese translation of the textbook in manuscript form that students can also use and that is available on request. The papers will be made available through the Web Learning system, and through a dedicated WeChat group.

2 Assessment

Assessment consists of:

- Mid-term exam: 30 points
- Final exam: 30 points
- Homework assignments: 30 points
- Presentation: 10 points

Homework assignments cover the logical materials. The philosophical papers will be covered by student presentations and classroom discussions.

The exams will test the students' abilities with regard to both the logical material as well as the philosophical material.

3 Homework

Answers to exercises need to be handed in before the start of class on the following Monday. This can be done through the Web Learning system . Homework will be graded and returned to the students by the Monday of the week after, before the start of class.

4 Presentations

Students are requested to sign up for a presentation at the start of the course. In case there are multiple people signing up for the same presentation, the teachers will assign at most three students to the same presentation. In such cases the students are assumed to coordinate their work by themselves.

Presentations should be not longer than 35 minutes. See below (section 7) for some guidelines for preparation of presentations.

Students are requested to email their (draft) presentations to the teacher by email, for comments and suggestions, at least 48 hours before the meeting in which the presentation is held.

5 Teachers

The course will be given by Martin Stokhof (email: m.j.b.stokhof@uva.nl; website: stokhof.org), and Wei Xintao (email: weixt22@mails.tsinghua.edu.cn), teaching assistant.

6 Schedule

The specification for each section contains:

- *Topic*:
  overview; introduction of core logical concepts and techniques
- *Textbook*:
  the chapter from the textbook in which the topics are explained
- *Exercises*:
  the relevant exercises
- *Theme*:
  the philosophical theme associated with those concepts and techniques
- *Textbook*:
  the chapter from the textbook in which the theme and the reading material are introduced
- *Reading*:
  the reading material that is related to the theme

Introduction

Section 1 (Sep 15 and 17): why logic?

- *Topic*:
  argument, argument scheme, validity; historical backgrounds; examples
- *Textbook*:
  chapter 1
- *Exercises*:
  n.a.
- *Theme*:
  usefulness and limitations of application of logic in philosophy
- *Textbook*:
  chapter 2
- *Reading*:
  Frege: preface from *Begriffsschrift*
  Russell: 'Logic as the Essence of Philosophy' in *Our Knowledge of the External World* (chapter 2, pp. 26–48)

Proposition logic

Section 2 (Sep 22 and 24): syntax

- *Topic*:
  truth functional connectives, truth tables; translation; syntax
- *Textbook*:
  chapter 3
- *Exercises*:
  1, 2, 3, 4, 5
- *Theme*:
  truth-functional, non-truth-functional and nonsensical connectives
- *Textbook*:
  chapter 4
- *Reading*:
  Prior: The Runabout Inference Ticket
  Belnap: Tonk, Plink and Plonk

Section 3 (Sep 29 and Oct 11): semantics

- *Topic*:
  valuations; equivalence; tautology, contradiction, contingency
- *Textbook*:
  chapter 5
- *Exercises*:
  1, 2, 3
- *Theme*:
  paradoxes of material implication: the limits of truth-functionality
- *Textbook*:
  chapter 6

- *Reading*:
  Suber: Paradoxes of Material Implication
  Sanford: introduction to *'If P then Q' Conditionals and the Foundations of Reasoning*

Section 4 (Oct 13 and 15): validity

- *Topic*:
  validity, counterexamples; functional completeness
- *Textbook*:
  chapter 7
- *Exercises*:
  1(b,i,m,o), 2(d,e), 3(c,e); 4, 5(a,d,g), 6, 7, 9(f,j,q), 10, 11, 12, 13
- *Theme*:
  logic and natural language: formal and informal consequence
- *Textbook*:
  chapter 8
- *Reading*:
  Harman, Chapters 1 and 2 of *Change in View*

Section 5 (Oct 20 and 22): natural deduction

- *Topic*:
  derivability and validity; introduction rules and elimination rules; conjunction, implication, disjunction
- *Textbook*:
  chapter 9
- *Exercises*:
  1, 2, 3, 4, 5
- *Theme*:
  deduction and meaning
- *Textbook*:
  chapter 10
- *Reading*:
  de Queiroz, On Reduction Rules, Meaning-as-use, and Proof-theoretic Semantics (first part)

Section 6 (Oct 27 and 29): natural deduction (cont.)

- *Topic*:
  negation
- *Textbook*:
  chapter 9
- *Exercises*:
  6, 7
- *Theme*:
  deduction and meaning
- *Textbook*:
  chapter 10

- *Reading*:
  de Queiroz, On Reduction Rules, Meaning-as-use, and Proof-theoretic Semantics (second part)

Midterm Exam

Section 7 (Nov 3 and 5): summary and midterm exam

- Nov 3: extra exercises and questions
- Nov 5: midterm exam

Predicate logic

Section 8 (Nov 10 and 12): syntax

- *Topic*:
  syntax: atomic sentences, domains, quantification; translation; scope and binding
- *Textbook*:
  chapter 11
- *Exercises*:
  1(a,b,d,g), 2(a,c,d), 3, 4
- *Theme*:
  ontology: quantification and existence; nominalism and universalism
- *Textbook*:
  chapter 12
- *Reading*:
  Quine: On What There Is

Section 9 (Nov 17 and 19): semantics; truth

- *Topic*:
  semantics: substitution method, assignments method
- *Textbook*:
  chapter 13
- *Exercises*:
  1, 2
- *Theme*:
  the nature of truth; object- and meta-language; self-reference and paradox
- *Textbook*:
  chapter 14
- *Reading*:
  Tarski: The Semantic of Conception of Truth

Section 10 (Nov 24 and 26): validity; identity

- *Topic 1*:
  validity and invalidity; proofs and counterexamples
- *Textbook*:
  chapter 15
- *Exercises*:
  4, 6(a,c); 9(a,d,f)
- *Topic 2*:
  identity: cardinals; descriptions
- *Textbook*:
  chapter 16
- *Exercises*:
  1(a), 2(a,d), 4(a,b(ii,iii)), 5(c,d), 6(a,d)

Section 11 (Dec 1 and Dec 3): relations and orderings

- *Topic*:
  properties of relations; orderings
- *Textbook*:
  chapter 18
- *Exercises*:
  1, 3, 4(a), 6(a,c), 7(b)
- *Theme*:
  descriptions and logically proper names; contextual analysis
- *Textbook*:
  chapter 17
- *Reading*:
  Russell: On Denoting

Section 12 (Dec 8 and Dec 10): natural deduction

- *Topic*:
  introduction and eliminations rules for quantifiers and identity
- *Textbook*:
  chapter 19
- *Exercises*:
  1(c), 2(b), 3(c), 4(a), 5(c)
- *Theme*:
  descriptions, use and utterance; presuppositional analysis
- *Textbook*:
  chapter 17
- *Reading*:
  Strawson: On Referring

Metalogic and Intensional Logic

Section 13 (Dec 15 and 17): soundness and completeness

- *Topic*:
  soundness and completeness of proposition logic
- *Textbook*:
  chapter 21
- *Exercises*:
  1(2), 2(4,8), 3
- *Theme*:
  logic, reasoning, normativity
- *Textbook*:
  chapter 20
- *Reading*:
  Engel: Logic, Reasoning and the Logical Constants

Section 14 (Dec 22 and 24): intensional logic

- *Topic*:
  intensional logic; relational semantics
- *Textbook*:
  chapter 24
- *Exercises*:
  1(a), 2(1,3), 3(1,2)
- *Theme*:
  sense and reference; rigid designation
- *Textbook*:
  chapter 25
- *Reading*:
  Frege: On Sense and Reference
  Kripke: Identity and Necessity

Final Exam

Section 15 (Dec 29 and Dec 31): summary and final exam

- Dec 29: extra exercises and questions
- Dec 31: final exam

7 Tips for presentations

Part of the requirements for the course is that you give a presentation of (part of) a philosophical paper. Here are a few tips to help you prepare a good presentation.

"Cut to the chase"

- philosophical papers are usually detailed, and, admittedly, sometimes long-winded
- your job is to distill the core views and main arguments, not to present each and every detail
- also keep in mind that your audience is supposed to have read the paper themselves

"It's your presentation, not that of the author"

- use your own words, not those of the author
- so, do not simply quote passages in which a particular point is made, but formulate that point in your own words
- quotations should only be used: if a. the point you want to make turns on a particular interpretation of the passage quoted; or b. the author has coined a phrase that is well-known and that summarises their views in a striking manner (Quine's "To be is to be the value of a variable" is a good example)
- and if you quote: always identify the quoted passage, by using quotation marks, or italics, or by otherwise marking it visually as such; and always state where the quoted passage comes from

"Present, don't read"

- just reading the material that is on your slides does not make for a very engaging presentation
- if there is a lot of material on a slide, especially if that is text that is written out in full, your audience will start reading your slides instead of listening to you
- so: make the material on your slides 'unreadable': do not use whole sentences, but short phrases that indicate a topic
- in that way you force yourself to speak freely and you force your audience to pay attention to what you are saying

"Mind the nitty gritty details"

- spellcheck and then spellcheck again; and be aware that not all spelling errors can be caught by a spellchecker
- be consistent in your use of capital letters, italics, bold face, quotation marks
- use italics for emphasis, but use it sparingly
- bold face is ok for headings, but in text is the typographical equivalent of shouting and should be avoided
- number your slides, so when someone has a question they can refer to what you state on a particular slide
- time your presentation by doing a 'dress rehearsal': it should be max. 20 minutes if you present part of the paper, max. 35 minutes if you are covering an entire paper

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